

Plan a retrosynthesis route (≤ 4 steps) for the given SMILES and output a reaction tree with purchasable starting materials.

`CCOC(=O)c1ccc(NC(=O)CN2CCOCC2)cc1`



User

Orchestrate Agent (Planning & Dispatch)

Planning

1. Analyze SMILES.
2. Generate retrosynthesis routes (< 4 steps).
3. Identify purchasable starting materials.
4. Compile reaction tree & report.

Dispatch

Task:

- ❑ Analyze the SMILES CCOC(=O)c1ccc(NC(=O)CN2CCOCC2)cc1 and generate a retrosynthesis route with no more than 4 steps.



Forge Agent

Forge Agent (Tool Calling & Execution)

Tool Call

Query:

CCOC(=O)c1ccc(NC(=O)CN2CCOCC2)cc1



DirectMultiStep (Generate routes explorer)



Tool Response

```
{
  "routes 1": "[\"... -> ...\"]",
  "routes 2": "[\"... -> ...\"]"
}
```



Analysis

- ✓ Route 1 selected (2 steps).
- ✓ Starting materials identified
- ✓ Reaction tree constructed.

Orchestrate Agent (Review & Output)

Review



- ✓ Routes generated
- ✓ successfully (2 steps ≤ 4).
- ✓ Purchasable materials confirmed.
- ✓ Reaction tree constructed.

Output



Retrosynthesis Report

Optimal Synthetic Route (2 steps):

- Step 1: Synthesis of Ethyl 4-(chloroacetyl)amino)benzoate
- Step 2: Final product formation

Purchasable Starting Materials:

1. Ethyl 4-aminobenzoate
2. Chloroacetyl chloride
3. Morpholine

Reaction Tree:

